

A locked structural rule explains WHO-graded rifampicin resistance in *Mycobacterium tuberculosis* rpoB and ranks clinical variants by drug-site proximity

rpoB slice of the TB drug-resistance structural map. Open public data only; all code, manifests and results in the project repository (rpoB/).

Abstract

Rifampicin (RIF) resistance in *M. tuberculosis* comes mostly from mutations in rpoB, the RNA polymerase beta subunit. The WHO 2023 mutation catalogue grades rpoB variants by statistical association with resistance, but a large share remain of uncertain significance. We asked whether one transparent, pre-registered structural rule - the minimum heavy-atom distance of the mutated residue to RIF in the *M. tuberculosis* RNAP-RIF crystal structure (PDB 5UHC), with a fixed 8.0 Å cutoff - is enough to explain WHO grade-1/2 entries, and whether it flags uncertain clinical variants as candidate resistance mutations. The rule and all gates were committed to version control before any outcome data were downloaded. A positive control confirmed the numbering map (S450 2.43 Å, H445 3.42 Å, D435 3.14 Å from RIF). The rule passed its validation gate: 123 of 136 grade-1/2 entries (90.4%, Wilson 95% CI 84.3-94.3%) lie within 8.0 Å, against 7.3% of grade-3 entries and 0 of 7 grade-4/5 protein-altering entries. The 13 misses all sit in the RRDR loop 8.4-12.1 Å from the drug (N438, P439, G442, L443, K446). Applying the rule to 5,286 CRyPTIC isolates whose only rpoB protein change was a single variant, no grade-3 or uncatalogued variant met the pre-registered novel-candidate gate (≥ 3 solo isolates, median MIC above the 0.5 mg/L ECOFF); four drug-site variants met the MIC criterion in single isolates only and are reported as thin. In an exploratory analysis, distance strongly tracked MIC across 53 variants with ≥ 3 solo isolates (Spearman $\rho = -0.80$, $p = 1.0e-12$), and five grade-1/2 contact-residue variants (L430P, H445N, H445S, D435G, S450V) showed median MICs at or below the ECOFF, matching the known borderline class. The result is a validated structural atlas of 28 resistance residues, a negative result for novel discovery that is preserved as such, and a quantitative link between drug-site geometry and resistance level.

Hypothesis

H1: WHO grade-1/2 rpoB RIF-resistance entries concentrate at residues within 8.0 Å of RIF in the *M. tuberculosis* RNAP-RIF structure, at a rate of at least 90%. H2: grade-3 or uncatalogued rpoB variants that meet the same structural criterion, and are the only rpoB protein change in at least three CRyPTIC isolates with median RIF MIC above the ECOFF, are candidate resistance mutations. H1 was the gate for any novel interpretation; H2 was evaluated only after H1 passed.

1. Introduction

Rifampicin is the backbone of first-line tuberculosis therapy, and resistance to it defines rifampicin-resistant and multidrug-resistant TB. RIF binds a pocket in the RNAP beta subunit next to the RNA exit channel and blocks extension of the nascent transcript beyond two to three nucleotides. Most

clinical resistance maps to an 81-bp "rifampicin resistance determining region" (RRDR, codons 426-452 in *M. tuberculosis* numbering; 507-533 in *E. coli*), which molecular diagnostics such as Xpert MTB/RIF target.

The WHO 2023 catalogue (second edition) grades each variant from 1 (associated with resistance) to 5 (not associated), using phenotype-genotype statistics on tens of thousands of isolates plus expert rules. Most *rpoB* rows are grade 3, uncertain significance. Structure-based approaches have been used before: Portelli et al. (2020) built machine-learning models from structural features to predict resistance beyond the RRDR; a 2022 Molecules study modelled clinical RpoB mutations against RIF binding; and structure-informed ML was revisited in 2024. Our aim differs in two ways. First, we test the simplest possible structural explanation, one distance cutoff fixed in advance, so its success or failure is directly interpretable and cannot be overfit. Second, we pre-register a success gate against the current WHO grades and a separate, stricter gate for novel claims on public clinical MIC data, and we report both honestly.

Prior-art verdict: CROWDED. Closest works: Portelli et al. 2020, Sci Rep 10:18120 (<https://www.nature.com/articles/s41598-020-74648-y>); "The Structural Basis of *M. tuberculosis* RpoB Drug-Resistant Clinical Mutations on Rifampicin Drug Binding", Molecules 2022 (<https://www.mdpi.com/1420-3049/27/3/885>); structure-informed ML preprint 2024 (<https://doi.org/10.1101/2024.08.15.608097>). Context: the *E. coli* RIF-site deep mutational scan, Nature 2023 (<https://www.nature.com/articles/s41586-023-06495-6>).

2. Methods

2.1 Pre-registration

The question, rule, thresholds and gates were written in GATES_LOCKED.md and committed as a standalone commit before any outcome data were fetched. A second standalone addendum, written before G1 was computed, defined how multi-residue in-frame entries are scored and recorded the G0 positive control. Git history in the repository independently shows gates, then data byte-lock, then addendum, then results. No threshold was changed after outcome data were seen.

2.2 Data

Structure: PDB 5UHC, the *M. tuberculosis* RNAP transcription initiation complex with RIF (ligand RFP 1201, bound to chain C, RpoB). Reference sequence: UniProt P9WGY9 (1178 aa); the WHO catalogue uses the 1172-aa H37Rv annotation, so WHO position = UniProt position - 6. Answer key: WHO 2023 catalogue master file (WHO-UCN-TB-2023.6), 1,502 *rpoB* rows for rifampicin. Clinical isolates: the CRyPTIC consortium release (Zenodo record 15680920), per-sample catalogue effects (EFFECTS table, 77,543 *rpoB*/RIF rows over 41,048 samples) and UKMYC5/6 RIF MICs (21,681 measurements). All files are listed with source URLs and SHA-256 checksums in *rpoB*/data/DATA_MANIFEST.md.

2.3 Structural rule

Hydrogens were removed and, for every resolved RpoB residue, the minimum distance between any residue heavy atom and any RIF heavy atom was computed with gemmi and numpy. The structure sequence was globally aligned to UniProt residues 7-1178 (Biopython PairwiseAligner); all 1,126 resolved residues mapped with 100% identity, and 5UHC author numbering equals WHO numbering + 6.

Residues at ≤ 4.5 Å are "direct contact", 4.5-8.0 Å "pocket shell" and > 8.0 Å "distal". A residue is "resistance-plausible" at ≤ 8.0 Å. In-frame deletions and delins are scored by the minimum over their span; insertions and duplications by their flanking or duplicated residues. Entries at unresolved residues count as failures.

2.4 Gates

G0 (positive control): S450, H445 and D435 must each lie within 6 Å of RIF. G1 (validation): at least 90% of WHO grade-1/2 rpoB RIF entries must be resistance-plausible. G1b (reported): the plausible rate in grade-4/5 entries must be lower. G2 (novel candidates): a grade-3 or uncatalogued variant that is resistance-plausible, is the only rpoB protein-altering variant in at least 3 phenotyped CRyPTIC isolates, and has median MIC above the ECOFF. Variants meeting the MIC criterion in fewer than three isolates are reported as thin.

2.5 CRyPTIC processing

Per-sample rpoB calls were parsed: missense calls (e.g. S450L) were converted to HGVS and matched to WHO grades, synonymous, promoter and null (X) calls were dropped, and nucleotide indels were mapped to codons. An isolate is "solo" for a variant when that variant is its only rpoB protein-altering call. MICs were taken from UKMYC plates; censored values were set to half the lowest ($\leq x$) or twice the highest ($> x$) concentration for medians. The ECOFF, 0.5 mg/L, is the highest MIC labelled S in the CRyPTIC binary phenotype (R begins at 1.0 mg/L). Isolates with no rpoB protein-altering call served as a wild-type reference.

2.6 Mechanism atlas and exploratory analysis

For each residue carrying a grade-1/2 entry we recorded the closest residue and RIF atoms, polar contacts within 3.5 Å, and which RIF moiety is closest. Moieties were defined from the bond graph of the RFP chemical component (naphthofuranone core, ansa bridge, hydrazone-piperazine tail). The distance-MIC correlation was not pre-registered and is labelled exploratory throughout.

3. Results

3.1 Positive control (G0): pass

The three canonical resistance residues sit in direct contact with RIF: S450 at 2.43 Å, H445 at 3.42 Å and D435 at 3.14 Å. Twenty-one RpoB residues are within 4.5 Å: 12 in the RRDR (S428-D435, H445, R448, S450, L452) and nine outside it (R167, V170, R459, P483, N487, I491, N604, R607, H674).

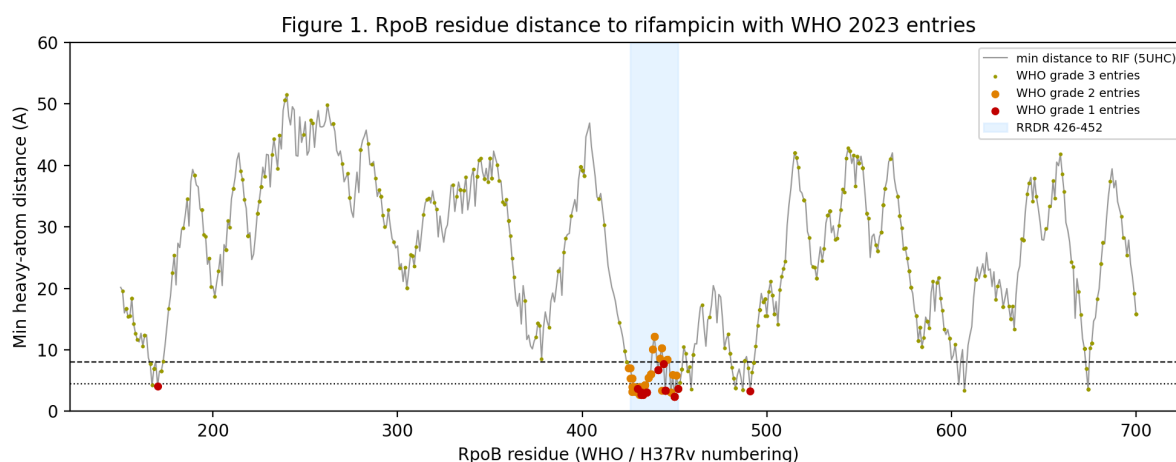


Figure 1. Minimum heavy-atom distance from each resolved RpoB residue (150-700 shown) to RIF in 5UHC. WHO grade-1 (red), grade-2 (orange) and grade-3 (olive) entries are plotted at their scored distance. Dashed line: locked 8.0 Å cutoff; dotted: 4.5 Å contact threshold. Blue band: RRDR.

3.2 Validation (G1): pass

Of 136 WHO grade-1/2 entries (103 missense, 27 in-frame deletions, 6 in-frame insertions), 123 (90.4%) were resistance-plausible: 89 direct contact, 34 pocket shell, 13 distal, none unresolved. The gate threshold was 90%, so the pass is narrow; a single additional miss would have failed it, and the lower bound of the 95% CI (84.3%) is below 90%. We report the pass as met but thin. The rule is selective: only 51 of 697 grade-3 protein-altering entries (7.3%) and 0 of 7 grade-4/5 protein-altering entries are within 8.0 Å (the other 552 grade-4/5 rows are synonymous or upstream).

Table 1. Structural class of WHO 2023 rpoB RIF entries by grade (grade-3 total includes 17 unresolved/unscorable).

WHO grade	Protein-altering entries	Contact ≤ 4.5 Å	Shell 4.5-8 Å	Distal >8 Å	Plausible (≤ 8 Å)
Grade 1-2	136	89	34	13	90.4%
Grade 3	697	23	28	629	7.3%
Grade 4-5	7	0	0	7	0.0%

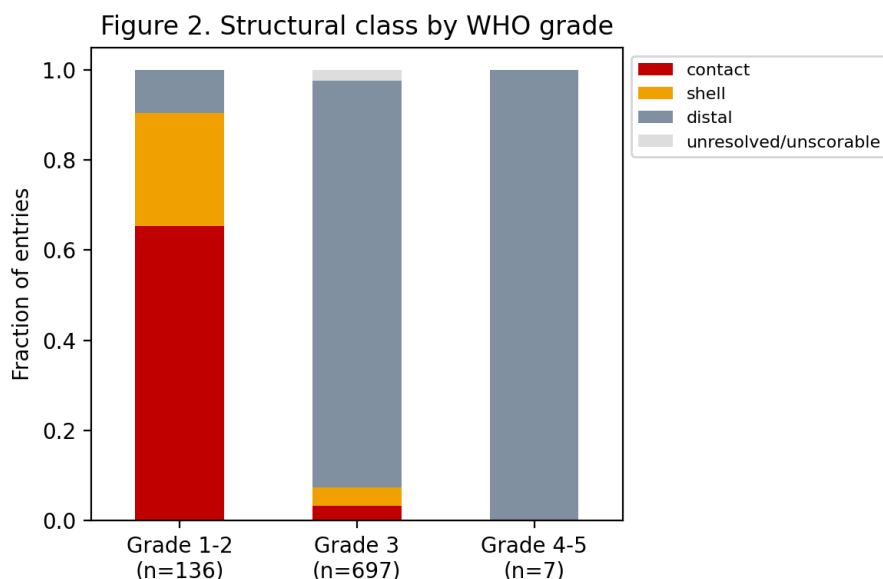


Figure 2. Fraction of WHO entries in each structural class, by grade.

3.3 Discrepancies

All 13 grade-1/2 entries outside 8.0 A fall at five RRDR residues, 8.4-12.1 A from RIF. None is far from the pocket in sequence; each sits in the loop that carries the contact residues D435 and H445. The most parsimonious explanation is second-shell effects on loop conformation, which a static distance rule cannot capture. These are hypotheses, not tested here, and the entries remain counted as failures.

Table 2. The 13 grade-1/2 entries not explained by the 8.0 A rule.

variant	grade	score	structural explanation (hypothesis)
p.Asn438del	2	10.09	RRDR loop; backbone links Q436/N437 shell to the P439 turn
p.Asn438His	2	10.09	RRDR loop; backbone links Q436/N437 shell to the P439 turn
p.Gly442Glu	2	8.64	glycine in loop; Glu adds bulk/charge next to shell residue S441
p.Leu443Phe	2	10.29	hydrophobic loop residue packing behind H445 and T444
p.Leu443Ser	2	10.29	hydrophobic loop residue packing behind H445 and T444
p.Leu443Trp	2	10.29	hydrophobic loop residue packing behind H445 and T444
p.Lys446Arg	2	8.42	lysine adjacent to H445; substitutions likely alter H445 side-chain placement
p.Lys446Gln	2	8.42	lysine adjacent to H445; substitutions likely alter H445 side-chain placement
p.Lys446Glu	2	8.42	lysine adjacent to H445; substitutions likely alter H445 side-chain placement
p.Lys446Thr	2	8.42	lysine adjacent to H445; substitutions likely alter H445 side-chain placement
p.Pro439Ala	2	12.12	proline turn of the RRDR loop; likely repositions the S431-D435 contact segment
p.Pro439Leu	2	12.12	proline turn of the RRDR loop; likely repositions the S431-D435 contact segment
p.Pro439Ser	2	12.12	proline turn of the RRDR loop; likely repositions the S431-D435 contact segment

3.4 Structural atlas

Table 3 lists every residue that carries a WHO grade-1/2 entry, with its distance, the closest RIF moiety, polar contacts and the number of CRyPTIC solo isolates. Resistance burden is extremely concentrated: S450 (3,111 solo isolates), D435 (840) and H445 (588) account for most clinical resistance, all in direct contact with RIF. The contact residues split between the naphthofuranone core (Q429, L430, Q432, R448, S450, L452) and the ansa bridge (S428, S431, F433, M434, D435, H445, I491), consistent with RIF being held by both faces of the pocket.

Table 3. rpoB rifampicin structural atlas (residues carrying WHO grade-1/2 entries). E. coli numbering = WHO + 81.

WHO pos	E. coli pos	aa	dist A	closest RIF moiety	polar contacts <=3.5 A	grade-1 /2 entries	CRyPTIC solo isolates	class
170	251	VAL	4.11	naphthofuranone core		1	16	direct contact
425	506	PHE	7.84	ansa bridge		1	0	pocket shell
426	507	GLY	7.09	ansa bridge		3	3	pocket shell
427	508	THR	5.33	ansa bridge		10	3	pocket shell
428	509	SER	3.98	ansa bridge		7	0	direct contact
429	510	GLN	3.22	naphthofuranone core	NE2-O5 3.36	5	0	direct contact
430	511	LEU	3.67	naphthofuranone core		4	100	direct contact
431	512	SER	3.97	ansa bridge		9	4	direct contact
432	513	GLN	3.07	naphthofuranone core	NE2-O2 3.07	15	47	direct contact
433	514	PHE	2.68	ansa bridge	N-O8 2.79; N-O9 3.46; O-O9 2.68	8	2	direct contact
434	515	MET	4.29	ansa bridge		14	2	direct contact
435	516	ASP	3.14	ansa bridge		18	840	direct contact
436	517	GLN	5.51	ansa bridge		9	1	pocket shell
437	518	ASN	6.04	ansa bridge		8	6	pocket shell
438	519	ASN	10.09	ansa bridge		3	0	distal/allosteric (RRDR loop)
439	520	PRO	12.12	ansa bridge		3	1	distal/allosteric (RRDR loop)
441	522	SER	6.74	ansa bridge		7	12	pocket shell
442	523	GLY	8.64	ansa bridge		1	0	distal/allosteric (RRDR loop)
443	524	LEU	10.29	ansa bridge		4	0	distal/allosteric (RRDR loop)
444	525	THR	7.77	ansa bridge		5	0	pocket shell
445	526	HIS	3.42	ansa bridge	NE2-O10 3.42	15	588	direct contact
446	527	LYS	8.42	ansa bridge		7	1	distal/allosteric (RRDR loop)
448	529	ARG	3.06	naphthofuranone core	NH1-O1 3.06; NH1-O10 3.24	3	1	direct contact
449	530	LEU	5.96	naphthofuranone core		1	0	pocket shell
450	531	SER	2.43	naphthofuranone core	OG-O2 2.43	11	3111	direct contact

WHO pos	E. coli pos	aa	dist A	closest RIF moiety	polar contacts <=3.5 A	grade-1 /2 entries	CRyPTI C solo isolates	class
451	532	ALA	5.89	naphthofuranone core		3	0	pocket shell
452	533	LEU	3.7	naphthofuranone core		4	102	direct contact
491	572	ILE	3.34	ansa bridge		1	62	direct contact

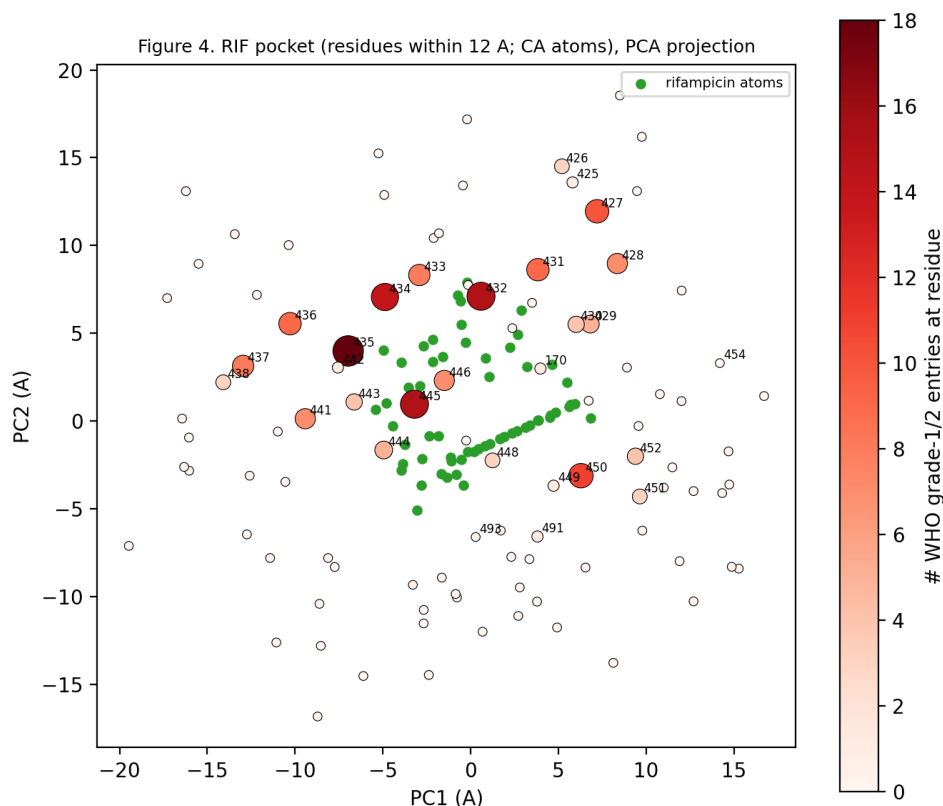


Figure 4. Two-dimensional principal-component projection of RIF heavy atoms (green) and CA atoms of residues within 12 Å. Circle size and colour show the number of WHO grade-1/2 entries at each residue.

3.5 Novel candidates (G2): none met the gate

We identified 5,286 phenotyped CRyPTIC isolates carrying exactly one *rpoB* protein-altering variant, covering 155 distinct variants (24 grade 1, 34 grade 2, 82 grade 3, 4 grade 4-5, 11 uncatalogued). The wild-type reference (5,470 isolates with no *rpoB* protein change) had median MIC 0.06 mg/L and 4.6% were called resistant, which sets the background for phenotype noise and resistance from outside *rpoB*. Twelve grade-3 or uncatalogued variants were resistance-plausible and phenotyped. None met G2. The three with at least three solo isolates (P454L, *n*=5; I491M, *n*=4; S493W, *n*=3) all had median MICs below the ECOFF. Four met the MIC criterion in one isolate each (I491Y, S493R, and premature stops D435* and L452*); the stop calls are implausible in an essential gene and more likely reflect mixed populations or calling artefacts. Under the locked rule this is a negative result for novel discovery, and we do not relax the gate.

Table 4. All grade-3/uncatalogued resistance-plausible variants with CRyPTIC solo isolates.

MUTATION	status	score	class	n_solo	median_mic	frac_R	G2 verdict
P454L	grade3	6.83	shell	5	0.03	0.0	below ECOFF
I491M	grade3	3.34	contact	4	0.09	0.25	below ECOFF
S493W	grade3	7.97	shell	3	0.06	0.0	below ECOFF
P454H	grade3	6.83	shell	2	0.075	0.0	below ECOFF
I491Y	grade3	3.34	contact	1	8.0	1.0	thin (n<3)
H445Z	uncatalogued	3.42	contact	1	0.015	0.0	below ECOFF
D435Z	uncatalogued	3.14	contact	1	1.0	1.0	thin (n<3)
L452Q	uncatalogued	3.7	contact	1	0.06	0.0	below ECOFF
L452Z	uncatalogued	3.7	contact	1	16.0	1.0	thin (n<3)
P454S	grade3	6.83	shell	1	0.06	0.0	below ECOFF
S493R	grade3	7.97	shell	1	16.0	1.0	thin (n<3)
V168M	grade3	6.98	shell	1	nan	0.0	no MIC

3.6 Exploratory: distance tracks resistance level

Across the 53 variants with at least three solo isolates, distance to RIF was strongly and inversely correlated with median MIC (Spearman rho = -0.80, p = 1.0e-12). Contact-residue variants had median MIC 8 mg/L and on average 72% of isolates resistant; distal variants had median MIC 0.06 mg/L and 1.6% resistant, matching wild type. This analysis was not pre-registered and uses CRyPTIC isolates that also contributed to the WHO catalogue, so it is not independent validation.

Within the contact class, five grade-1/2 variants had median MIC at or below the ECOFF: L430P (n=98, median 0.25, 15% R), H445N (n=42, 0.25, 24% R), H445S (n=7), D435G (n=4) and S450V (n=3). L430P, H445N and H445S are on the list of "disputed" (borderline) rpoB mutations that often test susceptible in MGIT despite conferring low-level resistance (J Clin Microbiol, <https://journals.asm.org/doi/10.1128/jcm.01599-17>: L430P, D435Y, H445C/L/N/S, L452P). D435G and S450V are not on that list; their low medians rest on 4 and 3 isolates and are thin. The structural rule places them in the pocket but does not predict their smaller effect, which points to a limit of distance-only models: the chemistry of the substitution matters as well as its location.

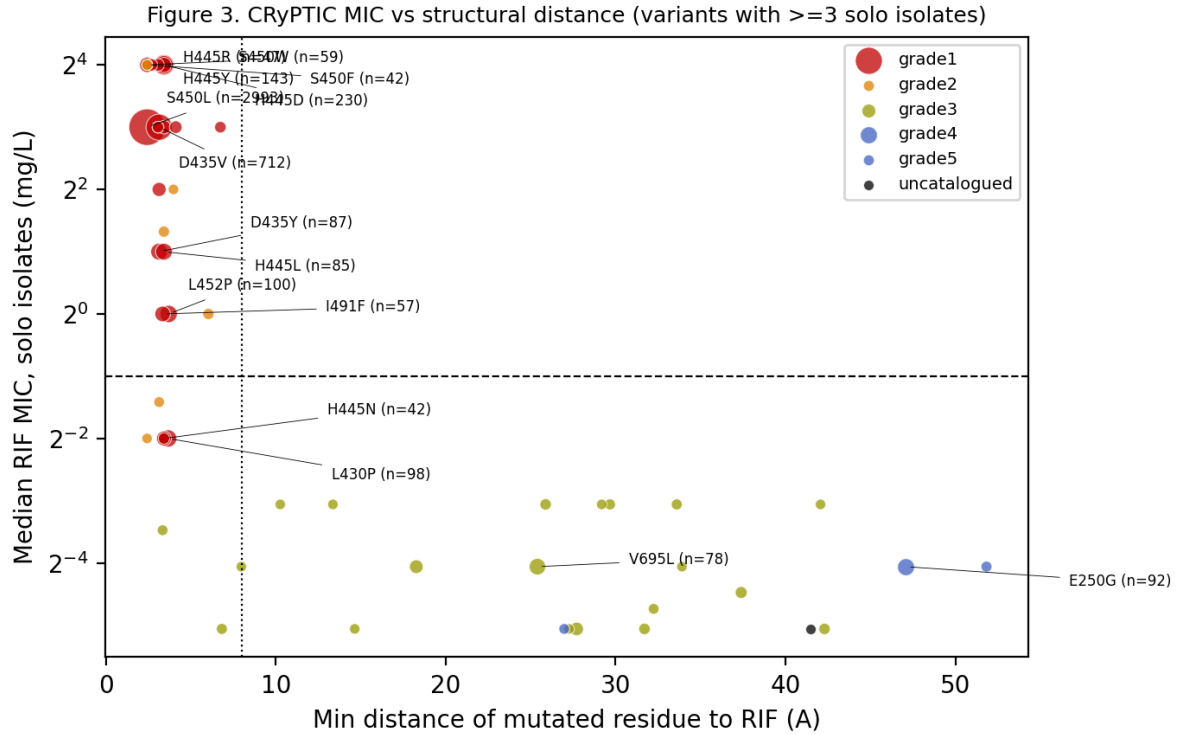


Figure 3. Median RIF MIC among solo isolates against the distance of the mutated residue to RIF, for variants with ≥ 3 solo isolates. Point size scales with isolate count. Dashed line: ECOFF 0.5 mg/L; dotted line: 8.0 A.

Table 5. CRyPTIC *rpoB* variants with ≥ 20 solo phenotyped isolates.

MUTATION	status	score	class	n_solo	n_high	median_mic	frac_R
S450L	grade1	2.43	contact	2993	2414	8.0	0.92
D435V	grade1	3.14	contact	712	612	8.0	0.85
H445D	grade1	3.42	contact	230	120	16.0	0.96
H445Y	grade1	3.42	contact	143	130	16.0	0.97
L452P	grade1	3.7	contact	100	71	1.0	0.51
L430P	grade1	3.67	contact	98	76	0.25	0.15
E250G	grade4	47.09	distal	92	89	0.06	0.01
D435Y	grade1	3.14	contact	87	58	2.0	0.77
H445L	grade1	3.42	contact	85	61	2.0	0.81
V695L	grade3	25.4	distal	78	66	0.06	0.0
S450W	grade1	2.43	contact	59	52	16.0	0.93
I491F	grade1	3.34	contact	57	34	1.0	0.65
H445R	grade1	3.42	contact	47	43	16.0	0.77
S450F	grade1	2.43	contact	42	39	16.0	0.9
H445N	grade1	3.42	contact	42	24	0.25	0.24
D435F	grade1	3.14	contact	31	25	4.0	0.84
V113I	grade3	18.27	distal	26	23	0.06	0.0
R791C	grade3	27.72	distal	24	22	0.03	0.0
H445C	grade1	3.42	contact	23	17	8.0	0.96

4. Discussion

One fixed distance cutoff on a single crystal structure explains just over 90% of the *rpoB* mutations that WHO grades as associated with RIF resistance, and very few of those it grades as uncertain or not associated. That supports the simple mechanism most often assumed for RIF resistance: mutations act by changing the drug-binding pocket directly. The rule also tells us what it cannot explain. Its failures cluster in one loop segment, and its successes include borderline mutations with near-wild-type MICs. Both point to effects on loop dynamics and on the chemistry of specific contacts, which distance alone does not capture.

The novel-discovery gate was not met. Under strict criteria, CRyPTIC contains no grade-3 variant at the drug site with enough solo isolates and high MICs to call it a resistance mutation. Part of the reason is circular: CRyPTIC isolates were part of the evidence base for the WHO 2023 grades, so a variant that is uncertain in the catalogue is likely to be rare or mixed in CRyPTIC too. Independent data (for example, later clinical collections not used by WHO) would be needed to test the thin candidates I491Y and S493R.

Practical use. The atlas (Table 3, `results/rpoB_structural_atlas.tsv`) and the per-residue distance table (`results/residue_distances.tsv`) form a small lookup tool: given any new *rpoB* variant in WHO numbering, it returns the structural class, the closest RIF moiety and the CRyPTIC MIC evidence. Our data suggest that a new variant at a contact residue deserves phenotypic follow-up, while a distal variant is very unlikely to raise the MIC.

Limitations. (1) One static structure, one ligand; no dynamics, no RIF analogues such as rifapentine or rifabutin. (2) The 8.0 Å cutoff was chosen a priori, not optimised; other cutoffs were deliberately not tested, to avoid fishing. (3) The G1 margin is narrow and its CI crosses 90%. (4) The grade-4/5 protein-altering set has only 7 entries. (5) The CRyPTIC data overlap the WHO training data. (6) Compensatory mutations in *rpoA*/*rpoC* and non-*rpoB* resistance are out of scope. (7) The distance-MIC analysis is exploratory.

5. Conclusion

A transparent, pre-registered structural rule passed its validation gate against WHO 2023 grades (123/136, 90.4%), produced a 28-residue mechanism atlas, and showed that distance to RIF tracks resistance level in public clinical data. It found no novel resistance mutations that pass a strict locked gate, and that negative result is reported as it stands.

Data and code availability

Repository `tb-resistance-structural-map`, directory `rpob/`: `code/00-06` scripts, `data/DATA_MANIFEST.md` (sources and SHA-256), `results/` tables and figures, `GATES_LOCKED.md` and `GATES_ADDENDUM_1.md`. Sources: RCSB PDB 5UHC; UniProt P9WGY9; WHO 2023 catalogue (github.com/GTB-tbsequencing/mutation-catalogue-2023); CRyPTIC (zenodo.org/records/15680920; ftp.ebi.ac.uk/pub/databases/cryptic).

References

World Health Organization. Catalogue of mutations in Mycobacterium tuberculosis complex and their association with drug resistance, 2nd ed. 2023. Master file: <https://github.com/GTB-tbsequencing/mutation-catalogue-2023>

CRyPTIC Consortium. A data compendium associating the genomes of 12,289 Mycobacterium tuberculosis isolates with quantitative resistance phenotypes to 13 antibiotics. PLoS Biol 2022. <https://doi.org/10.1371/journal.pbio.3001721>

RCSB PDB 5UHC: Crystal structure of Mycobacterium tuberculosis transcription initiation complex with rifampicin. <https://www.rcsb.org/structure/5UHC>

Portelli S. et al. Prediction of rifampicin resistance beyond the RRDR using structure-based machine learning approaches. Sci Rep 2020. <https://www.nature.com/articles/s41598-020-74648-y>

The Structural Basis of Mycobacterium tuberculosis RpoB Drug-Resistant Clinical Mutations on Rifampicin Drug Binding. Molecules 2022, 27(3):885. <https://www.mdpi.com/1420-3049/27/3/885>

Predicting rifampicin resistance in M. tuberculosis using machine learning informed by protein structural and chemical features. bioRxiv 2024. <https://doi.org/10.1101/2024.08.15.608097>

Role of disputed mutations in the rpoB gene in interpretation of automated liquid MGIT culture results for rifampin susceptibility testing of Mycobacterium tuberculosis. J Clin Microbiol (DOI 10.1128/JCM.01599-17). <https://journals.asm.org/doi/10.1128/jcm.01599-17>

High-resolution landscape of an antibiotic binding site. Nature 2023. <https://www.nature.com/articles/s41586-023-06495-6>

Appendix A. All WHO 2023 grade-1/2 rpoB RIF entries with structural score

variant	grade	effect	score	class
p.Ala451Gly	2	missense_variant	5.89	shell
p.Ala451Val	2	missense_variant	5.89	shell
p.Arg448Gln	2	missense_variant	3.06	contact
p.Arg448Leu	2	missense_variant	3.06	contact
p.Arg448Lys	2	missense_variant	3.06	contact
p.Asn437_Asn438del	2	inframe_deletion	6.04	shell
p.Asn437Asp	2	missense_variant	6.04	shell
p.Asn437His	2	missense_variant	6.04	shell
p.Asn437Ile	2	missense_variant	6.04	shell
p.Asn437Ser	2	missense_variant	6.04	shell
p.Asn437Tyr	2	missense_variant	6.04	shell
p.Asn438del	2	inframe_deletion	10.09	distal
p.Asn438His	2	missense_variant	10.09	distal
p.Asp435_Gln436del	2	inframe_deletion	3.14	contact
p.Asp435_Gln436delinsGlu	2	inframe_deletion	3.14	contact
p.Asp435Ala	2	missense_variant	3.14	contact
p.Asp435Asn	2	missense_variant	3.14	contact
p.Asp435del	2	inframe_deletion	3.14	contact
p.Asp435Glu	2	missense_variant	3.14	contact
p.Asp435Gly	2	missense_variant	3.14	contact
p.Asp435His	2	missense_variant	3.14	contact
p.Asp435Leu	2	missense_variant	3.14	contact
p.Asp435Phe	1	missense_variant	3.14	contact
p.Asp435Tyr	1	missense_variant	3.14	contact
p.Asp435Val	1	missense_variant	3.14	contact
p.Gln429His	2	missense_variant	3.22	contact
p.Gln429Leu	2	missense_variant	3.22	contact
p.Gln429Pro	2	missense_variant	3.22	contact
p.Gln432_Asp435delinsHis	1	inframe_deletion	2.68	contact
p.Gln432_Met434del	2	inframe_deletion	2.68	contact
p.Gln432_Met434delinsLeu	2	inframe_deletion	2.68	contact
p.Gln432_Phe433del	2	inframe_deletion	2.68	contact
p.Gln432Asn	2	missense_variant	3.07	contact
p.Gln432del	2	inframe_deletion	3.07	contact
p.Gln432Glu	2	missense_variant	3.07	contact
p.Gln432His	2	missense_variant	3.07	contact
p.Gln432Leu	1	missense_variant	3.07	contact
p.Gln432Lys	1	missense_variant	3.07	contact
p.Gln432Pro	1	missense_variant	3.07	contact

variant	grade	effect	score	class
p.Gln436_Asn437del	2	inframe_deletion	5.51	shell
p.Gln436Arg	2	missense_variant	5.51	shell
p.Gln436Asn	2	missense_variant	5.51	shell
p.Gln436del	2	inframe_deletion	5.51	shell
p.Gln436Pro	2	missense_variant	5.51	shell
p.Gly426_Thr427del	2	inframe_deletion	5.33	shell
p.Gly426Ser	2	missense_variant	7.09	shell
p.Gly442Glu	2	missense_variant	8.64	distal
p.His445_Lys446del	2	inframe_deletion	3.42	contact
p.His445_Lys446delinsGln	2	inframe_deletion	3.42	contact
p.His445Arg	1	missense_variant	3.42	contact
p.His445Asn	1	missense_variant	3.42	contact
p.His445Asp	1	missense_variant	3.42	contact
p.His445Cys	1	missense_variant	3.42	contact
p.His445Gln	2	missense_variant	3.42	contact
p.His445Gly	2	missense_variant	3.42	contact
p.His445Leu	1	missense_variant	3.42	contact
p.His445Phe	2	missense_variant	3.42	contact
p.His445Pro	2	missense_variant	3.42	contact
p.His445Ser	1	missense_variant	3.42	contact
p.His445Thr	2	missense_variant	3.42	contact
p.His445Tyr	1	missense_variant	3.42	contact
p.Ile491Phe	1	missense_variant	3.34	contact
p.Leu430_Gln432del	2	inframe_deletion	3.07	contact
p.Leu430_Ser431insArg	2	inframe_insertion	3.67	contact
p.Leu430Arg	2	missense_variant	3.67	contact
p.Leu430Pro	1	missense_variant	3.67	contact
p.Leu443_Lys446delinsProGln	2	inframe_deletion	3.42	contact
p.Leu443Phe	2	missense_variant	10.29	distal
p.Leu443Ser	2	missense_variant	10.29	distal
p.Leu443Trp	2	missense_variant	10.29	distal
p.Leu449Met	2	missense_variant	5.96	shell
p.Leu452Met	2	missense_variant	3.7	contact
p.Leu452Pro	1	missense_variant	3.7	contact
p.Leu452Val	2	missense_variant	3.7	contact
p.Lys446Arg	2	missense_variant	8.42	distal
p.Lys446Gln	2	missense_variant	8.42	distal
p.Lys446Glu	2	missense_variant	8.42	distal
p.Lys446Thr	2	missense_variant	8.42	distal
p.Met434_Asn437delinsIle	2	inframe_deletion	3.14	contact
p.Met434_Asp435del	2	inframe_deletion	3.14	contact
p.Met434_Asp435insVal	2	inframe_insertion	3.14	contact
p.Met434Arg	2	missense_variant	4.29	contact

variant	grade	effect	score	class
p.Met434Ile	2	missense_variant	4.29	contact
p.Met434Leu	2	missense_variant	4.29	contact
p.Met434Thr	2	missense_variant	4.29	contact
p.Met434Val	2	missense_variant	4.29	contact
p.Phe425_Gly426del	2	inframe_deletion	7.09	shell
p.Phe433_Asp435del	2	inframe_deletion	2.68	contact
p.Phe433_Gln436del	2	inframe_deletion	2.68	contact
p.Phe433dup	1	inframe_insertion	2.68	contact
p.Pro439Ala	2	missense_variant	12.12	distal
p.Pro439Leu	2	missense_variant	12.12	distal
p.Pro439Ser	2	missense_variant	12.12	distal
p.Ser428Arg	2	missense_variant	3.98	contact
p.Ser428Gly	2	missense_variant	3.98	contact
p.Ser428Ile	2	missense_variant	3.98	contact
p.Ser428Thr	2	missense_variant	3.98	contact
p.Ser431_Gln432insArg	2	inframe_insertion	3.07	contact
p.Ser431_Gln432insHis	2	inframe_insertion	3.07	contact
p.Ser431_Met434del	2	inframe_deletion	2.68	contact
p.Ser431Arg	2	missense_variant	3.97	contact
p.Ser431Asn	2	missense_variant	3.97	contact
p.Ser431Gly	2	missense_variant	3.97	contact
p.Ser431Thr	2	missense_variant	3.97	contact
p.Ser441Ala	2	missense_variant	6.74	shell
p.Ser441Gln	1	missense_variant	6.74	shell
p.Ser441Leu	1	missense_variant	6.74	shell
p.Ser441Lys	2	missense_variant	6.74	shell
p.Ser441Met	2	missense_variant	6.74	shell
p.Ser441Trp	2	missense_variant	6.74	shell
p.Ser441Val	2	missense_variant	6.74	shell
p.Ser450_Leu452del	2	inframe_deletion	2.43	contact
p.Ser450Ala	2	missense_variant	2.43	contact
p.Ser450Cys	2	missense_variant	2.43	contact
p.Ser450Gln	2	missense_variant	2.43	contact
p.Ser450Gly	2	missense_variant	2.43	contact
p.Ser450Leu	1	missense_variant	2.43	contact
p.Ser450Met	2	missense_variant	2.43	contact
p.Ser450Phe	1	missense_variant	2.43	contact
p.Ser450Trp	1	missense_variant	2.43	contact
p.Ser450Tyr	1	missense_variant	2.43	contact
p.Ser450Val	2	missense_variant	2.43	contact
p.Thr427_Gln429del	2	inframe_deletion	3.22	contact
p.Thr427_Gln429delinsLys	2	inframe_deletion	3.22	contact
p.Thr427_Ser428del	2	inframe_deletion	3.98	contact

variant	grade	effect	score	class
p.Thr427Ala	2	missense_variant	5.33	shell
p.Thr427Asn	2	missense_variant	5.33	shell
p.Thr427Gly	2	missense_variant	5.33	shell
p.Thr427Ile	2	missense_variant	5.33	shell
p.Thr427Pro	2	missense_variant	5.33	shell
p.Thr427Ser	2	missense_variant	5.33	shell
p.Thr444dup	1	inframe_insertion	7.77	shell
p.Thr444Ile	2	missense_variant	7.77	shell
p.Thr444Pro	2	missense_variant	7.77	shell
p.Thr444Ser	2	missense_variant	7.77	shell
p.Val170Phe	1	missense_variant	4.11	contact

Appendix B. Gate chronology (git)

af62766 2026-09-23 13:49:12 +0530 builder 8: rpoB paper draft (15 pp PDF), README, payload MANIFEST.sha256

0356251 2026-09-23 13:47:00 +0530 builder 8: rpoB structural atlas (CCD-derived RIF moieties), exploratory distance-MIC analysis, figures 1-4

c77e75d 2026-09-23 13:45:33 +0530 builder 8: rpoB G2 result - 0 novel candidates meet locked gate (4 thin, n=1); CRyPTIC solo-isolate MIC table

153ab64 2026-09-23 13:44:01 +0530 builder 8: rpoB G1 validation result - PASS 123/136 (90.4%) vs WHO 2023 grade-1/2; 13 discrepancies documented

2e8dd5f 2026-09-23 13:43:38 +0530 builder 8: rpoB gates addendum 1 (entry-scoring definitions, G0 positive control) before G1 computed

b28e663 2026-09-23 13:42:35 +0530 builder 8: rpoB data byte-lock (sources, derived subsets, sha256 manifest; bulk raw gitignored with URL+sha256)

4cce4ec 2026-09-23 13:36:20 +0530 builder 8: LOCKED GATES for rpoB slice - pre-registered before any outcome data (originally committed locally 2026-09-23T13:36:20+05:30 as ababda6)